PNAS RESEARCH ARTICLE | NEUROSCIENCE

Energy prioritization and neurometabolic scaling in a social insect brain

Zach N. Coto[a,1], Dajia Ye[b], Sara Arganda[c,d], Jon F. Harrison[e], and James F. A. Traniello[a]

Affiliations are included on p. 10.

Edited by Raghavendra Gadagkar, Indian Institute of Science, Bangalore, India; received February 18, 2026; accepted June 17, 2026

Brains are metabolically costly and due to their high energetic demands may receive priority under conditions of reduced energy availability. Such preferential energy allocation to the brain has been described for humans and other mammals, but previous studies have not directly quantified brain metabolic rate or simultaneously considered whole-body metabolism. We recorded brain metabolic rate ex vivo, brain mass, body mass, and whole-body metabolic rate in workers of the ant *Tetramorium immigrans* and found the proportion of whole-body energy allocated to the brain nearly doubled to 45% when nutritionally stressed. Body metabolic rate was significantly reduced without a comparable decrease in brain metabolic rate. The ability of nutritionally compromised workers to recognize and aggressively respond to a sympatric competitor was not affected, indicating that sensory perception and neural processing necessary for this critical behavior are maintained during energy limitation. Our finding that similar patterns of brain energy prioritization occur across remotely related clades with exceptionally different body sizes, respiratory systems, and brain allometries suggests conservation of ancient neurohormonal mechanisms or functional convergence of processes to protect the brain. Furthermore, although the worker brain was 5% of body mass, brain metabolic rate was 24 to 30% of body metabolic rate under unstressed conditions, remarkably similar to the pattern of humans and other mammals. Brain metabolic rate of *T. immigrans* was predictable from the scaling of brain metabolic rate in mammals, suggesting ecological, physiological, and evolutionary effects of body size on brain metabolic rate are common.

brain evolution | expensive tissue | miniaturization | brain metabolism | social brain

Significance

It has been hypothesized that brain physiology has evolved to prioritize energy allocation during nutritional stress, but brain and whole-body metabolism have not been measured in combination. We examined the effect of nutritional stress on whole-body and brain metabolic rate ex vivo and performance of a critical and complex task in the ant *Tetramorium immigrans* . Nutritional stress approximately doubled the fraction of energy consumed by the brain, and performance of a key defensive task was unimpaired. Our study demonstrates the brain receives priority to satisfy energy requirements and provide behavioral resilience during nutritional stress. Brain energy prioritization and patterns of neurometabolic scaling may be broadly conserved across animal taxa.

The human brain requires 20% of whole-body energy despite comprising only 2% of body mass (1–4). This high neurometabolic expenditure is hypothesized to result in evolutionary trade-offs in investment in other organ systems (5–10). The exceptional neurometabolic requirement has been hypothesized to have selected for preferential access to energy by the brain during physiological stress (11, 12). Diverse mechanisms have evolved to maintain brain function and behavioral performance under fitness– and survival-compromising conditions (13). Although the theory of brain energy prioritization was developed in the context of how the human brain responds to stress, it has the potential to broadly explain adaptive characteristics of brain physiology and potential tradeoffs with other energetically costly systems across animal clades. Species varying widely in brain size and behavioral and/or cognitive challenges may also differ in brain metabolic demands and therefore neurometabolic responses to stress. However, prior tests of the hypothesis of brain energy prioritization during nutritional stress have been indirect and therefore inconclusive.

Evaluating neurometabolic prioritization theory has been restricted by the availability of technical methods to directly measure brain and body metabolic rates under physiological stress. Previous studies of the effects of physiological stress on neurometabolism have not directly measured metabolic rates but instead have focused on brain mass or variation in the concentration of energy metabolites or hormones such as insulin (12). These results may simply reflect shifts in substrate use without changes in metabolic demands (14–20). Otherwise, brain metabolic rates have been estimated in humans (3, 21, 22), several mammals (4, 23, 24), one species of bird (25), and several insects (26–32), but do not assess metabolic impacts of stress. Studies of tradeoffs between the brain and other systems found negative correlations between memory and survival or starvation resistance (33, 34), reproduction (35–37), and physical performance (38), but have not directly recorded brain metabolic rate and therefore do not demonstrate brain energy prioritization. A strong test of the hypothesis of neurometabolic prioritization during nutritional stress requires that brain and body metabolic rate be jointly assessed.

Author contributions: Z.N.C., D.Y., S.A., J.F.H., and J.F.A.T. designed research; Z.N.C., D.Y., and S.A. performed research; Z.N.C., D.Y., S.A., and J.F.H. analyzed data; and Z.N.C., D.Y., S.A., J.F.H., and J.F.A.T. wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission.

Copyright © 2026 the Author(s). Published by PNAS. This article is distributed under Creative Commons Attribution-NonCommercial-NoDerivatives License 4.0 (CC BY-NC-ND).

[1]To whom correspondence may be addressed. Email: zcoto@bu.edu.

This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2605431123/-/DCSupplemental.

Published July 27, 2026.

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

Ants provide a unique and powerful model system for testing hypotheses of brain energy prioritization as the metabolic rates of whole insect brains can be measured ex vivo (29, 30, 32), large numbers of individuals can be assayed and nutrition can be readily manipulated. Additionally, as social insects, ants provide a powerful test of the generality of brain energy use and prioritization as they are only distantly related to the mammals that have been previously studied, are miniature in comparison and have significantly different respiratory physiology utilizing a tracheal system for gas exchange. The origin of societies—one of the major transitions in the history of life (39)—was accompanied by adaptive changes in brain size and its metabolic consequences to the behavioral challenges of group living (40, 41). Eusocial insects live in remarkably cooperative groups in which behavioral performance and/or cognitive demands on individual colony members may be reduced due to collective intelligence and division of labor, potentially decreasing worker brain energy costs (40, 42–44). Group decision-making in insect societies may lower brain energy usage, for example, in the mushroom bodies, centers of higher-order processing that regulate social behavior (45). Additionally, social insects exhibit significantly smaller brains than vertebrates (46, 47) and, although invertebrates may exhibit lower information-processing efficiency than vertebrates (48), reduced neuron size is expected to reduce action potential costs (49, 50), Therefore, social insect brains may have lower energy costs than vertebrates and be able to maintain neural function under stress without prioritizing energy allocation to the brain.

Alternatively, the smaller brain size of social insects relative to vertebrates may be associated with increased mass-specific metabolic rates (51, 52), a pattern of body metabolic scaling shared by both clades (53–56). This suggests brain energy prioritization may be as critical to meet brain metabolic demands in social insects as in vertebrates. Insects and vertebrates also share homologous biochemical pathways to fuel neurometabolism (30) and centers of higher-order processing have ancient genetic homology (57). Therefore, common mechanisms may have evolved to allow brain energy prioritization across both clades. Despite the significant impacts of sociality on insect brain evolution, neurophysiological systems underpinning brain prioritization and neurometabolic scaling may be conserved.

Here, we ask: Do social insect brains have energy priority under stress? Are their brains, like those of mammals, disproportionately costly relative to their size? To address these questions, we measured metabolic rates in brains ex vivo and recorded whole body metabolic rates in workers of the ant *Tetramorium immigrans* fed ad libitum or that were nutritionally compromised. We hypothesized that nutritional stress due to food deprivation will reduce body metabolic rate without impacting brain metabolic rate, resulting in an increase in the proportion of energy allocated to the brain. Additionally, we expected neuroenergetic prioritization will enable the maintenance of critical behavioral functions requiring sensory perception, information processing, and effective motor response (58–61). Therefore, we hypothesize that worker aggressive behavior, which requires sensory recognition of threats to colony security from significant sympatric competitors as well as appropriate motor responses for aggressive behavior, will be preserved during nutritional stress. Collaterally, general locomotory output, which is less critical to colony defense, may be reduced in association with reduced body metabolic rate. We address these questions in the broad evolutionary context of neurometabolic scaling across phylogenetically distant taxa.

Results

We recorded data from workers cultured in subcolonies (N = 30 workers per subcolony) fed ad libitum (controls), and workers placed in same-size subcolonies without food (food-deprived). Because social processes may regulate worker foraging and nutritional intake (62), we also measured body and brain metabolic rates from workers within several hours of removal from colonies with fecund queens (queenright colonies). This allowed us to identify the metabolic effects of workers cultured in queenless subcolonies and the impact of an altered social environment (details in *Materials and Methods*). Because some prior studies (53, 63, 64) have shown strong effects of diet on body or brain metabolism, additionally, we estimate the potential role of feeding ecology on variation in brain and body metabolic rate. Workers in each treatment were sampled from environments expected to differ in food quality and availability: laboratory colonies (lab colonies), colonies from the Boston (urban colonies), and colonies inhabiting open mixed-vegetation fields (field colonies). Data on brain mass, body mass, brain metabolic rates, and body metabolic rates were recorded. Because metabolic rate is known to scale with size, we not only used brain and body mass as fixed effects in our statistical models (detail provided in *Materials and Methods*) but also provide brain mass-specific metabolic rate (brain metabolic rate divided by brain mass) and body mass–specific metabolic rate (body metabolic rate divided by body mass) as intuitive metrics for comparing metabolic rates across workers that may vary in brain and body size. Additionally, to evaluate the proportion of body mass and body energy that are allocated to the brain, we provide relative brain mass (brain mass divided by body mass) and relative brain metabolic rate (brain metabolic rate divided by body metabolic rate). The averages and SD of these data are provided in *SI Appendix*, Table S1.

Worker brain mass was not significantly impacted by treatment (Fig. 1*A* and Table 1, $\chi^2 = 0.87$, $P = 0.65$), but there was a significant treatment-body mass interaction effect on brain mass (Table 1, $\chi^2 = 7.09$, $P = 0.03$). This significant interaction occurred because only workers from queenright colonies exhibited significant scaling of brain mass to body mass (*SI Appendix*, Fig. S13, slope = 1.07, $r^2 = 0.20$, $P < 0.01$). Body mass was significantly affected by treatment (Fig. 1*B* and Table 1, $F = 30.48$, $P < 0.01$); controls had higher body masses than food-deprived workers (Fig. 1*B* and Table 2, $T = 7.43$, $P < 0.01$) and workers from queenright colonies (Fig. 1*B* and Table 2, $T = 5.66$, $P < 0.01$).

We found no significant treatment effect on brain metabolic rate (Fig. 2*A* and Table 1, $F = 1.38$, $P = 0.25$), but there was a significant treatment–environment interaction (*SI Appendix*, Fig. S5 and Table 1, $F = 3.88$, $P = 0.01$), with queenright workers collected from field colonies exhibiting significantly higher brain metabolic rates than food-deprived workers from field colonies (*SI Appendix*, Fig. S5 and Table S2, $T = 3.32$, $P < 0.01$), without differences between these treatments in other environments (*SI Appendix*, Table S2). Similarly, we found no significant treatment effect on brain mass-specific metabolic rate (Fig. 3*A* and Table 1, $F = 1.20$, $P = 0.31$), but there was a significant treatment–environment interaction effect (*SI Appendix*, Fig. S6 and Table 1, $F = 6.31$, $P < 0.01$), with queenright workers collected from field colonies exhibiting significantly higher brain mass-specific metabolic rates than food-deprived workers from field colonies (*SI Appendix*, Fig. S6 and Table S3, $T = 4.57$, $P < 0.01$) and controls from field colonies (*SI Appendix*, Fig. S6 and Table S3, $T = 3.84$, $P < 0.01$). Queenright workers from lab colonies also

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

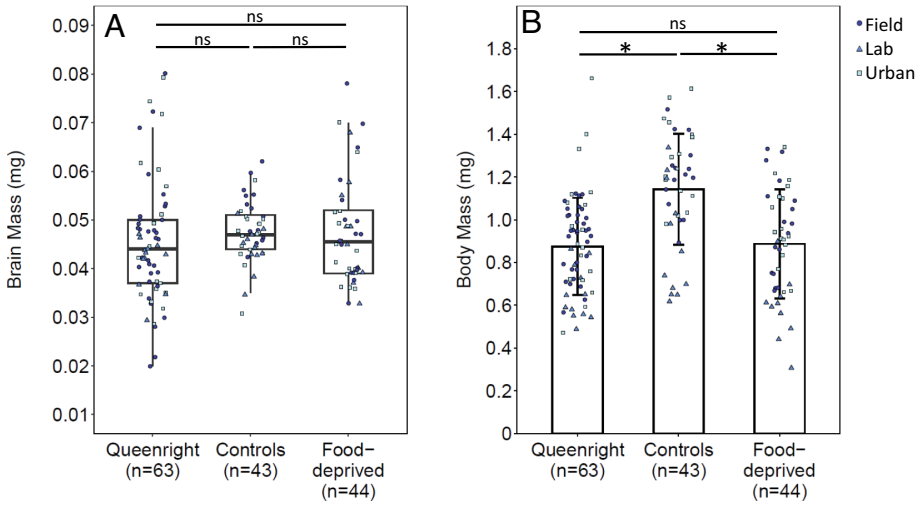


Fig. 1. (*A*) Worker brain mass was not affected by treatment (Table 1, $\chi^2 = 0.87$, $P = 0.65$). The colony-collection environment did not have a significant effect (Table 1, $\chi^2 = 1.50$, $P = 0.47$). Box plots represent medians ± 150% of the interquartile range. (*B*) Worker body mass was significantly affected by treatment (Table 1, F = 30.48, $P < 0.01$), but not colony environment (Table 1, F = 2.14, $P = 0.16$). Controls had significantly higher body mass than queenright workers (Table 2, T = 5.66, $P < 0.01$) and food-deprived workers (Table 2, T = 7.43, $P < 0.01$). Data represent means ± SD.

had significantly higher brain mass-specific metabolic rates than controls from lab colonies (*SI Appendix*, Fig. S6 and Table S3, T = 2.51, $P = 0.04$).

Significant treatment effects were found for body metabolic rate (Fig. 2*B* and Table 1, F = 10.24, $P < 0.01$) and body mass–specific metabolic rate) Fig. 3*B* and Table 1, F = 19.21, $P < 0.01$). Food-deprived workers had significantly lower body metabolic rates than controls (Fig. 2*B* and Table 2, T = 3.86, $P < 0.01$) and queenright workers (Fig. 2*B* and Table 2, T = 3.88, $P < 0.01$). Food-deprived workers also had lower body mass–specific metabolic rates than controls (Fig. 3*B* and Table 2, T = 4.11, $P < 0.01$) and queenright workers (Fig. 3*B* and Table 2, T = 5.98, $P < 0.01$). Notably, while food-deprived workers exhibited a nonsignificant 5% decrease in brain mass-specific metabolic rate from controls, they showed a significant 30% decrease in body mass–specific metabolic rate from controls (*SI Appendix*, Table S1). There was no effect of colony-collection environment on body metabolic rate (*SI Appendix*, Fig. S7 and Table 1, F = 0.03, $P = 0.97$) or body mass–specific metabolic rate (*SI Appendix*, Fig. S8 and Table 1, F = 0.96, $P = 0.44$).

Worker brain mass was 4 to 6% of body mass (*SI Appendix*, Table S1). Relative brain mass was affected by treatment (Fig. 3*B* and Table 1, $\chi^2 = 33.25$, $P < 0.01$) and colony-collection environment (Fig. 4*A* and Table 1, $\chi^2 = 30.63$, $P < 0.01$). Relative brain mass was lower for controls than food-deprived workers (Fig. 3*B* and Table 2, Z = 4.72, $P < 0.01$) and workers from queenright colonies (Fig. 4*A* and Table 2, Z = 4.80, $P < 0.01$). There was no significant treatment/environment interaction effect for brain mass (*SI Appendix*, Fig. S1 and Table S5, $\chi^2 = 4.72$, $P = 0.32$), body mass (*SI Appendix*, Fig. S2 and Table S6, F = 0.46, $P = 0.77$), or relative brain mass (*SI Appendix*, Fig. S3 and Table S7, $\chi^2 = 7.33$, $P = 0.12$).

We found a significant treatment effect on relative brain metabolic rate (Fig. 4*B* and Table 1, F = 10.94, $P < 0.01$). Brain metabolic rates of food-deprived workers were approximately 45% of body metabolic rate (*SI Appendix*, Table S1), significantly greater than relative brain metabolic rates of controls (*SI Appendix*, Table S1, 24%; Fig. 2*B* and Table 2, F = 3.79, $P < 0.01$) and workers from queenright colonies (*SI Appendix*, Table S1, 30%; Fig. 2*B* and *SI Appendix*, Table S3, F = 2.87, $P = 0.01$). The colony-collection environment did not have a significant effect on relative brain metabolic rate (*SI Appendix*, Fig. S4 and Table 1, F = 0.20, $P = 0.82$).

Worker movement within respirometry chambers was significantly affected by treatment (Fig. 5 and Table 1, $\chi^2 = 32.57$, $P < 0.01$). Queenright workers moved significantly more than food-deprived workers (Fig. 5 and Table 2, Z = 4.42, $P < 0.01$) and controls (Fig. 5 and Table 2, Z = 5.01, $P < 0.01$). Additionally, the colony-collection environment influenced worker movement (*SI Appendix*, Fig. S10 and Table 2, Z = 8.76, $P = 0.01$) workers from field colonies exhibited greater movement than workers from urban colonies (*SI Appendix*, Fig. S10 and Table 2, Z = 2.92, $P < 0.01$). There were no treatment/environment interaction effects on movement (*SI Appendix*, Fig. S10 and Table 1, $\chi^2 = 2.63$, $P = 0.62$).

Worker behavioral responses to encounters with the ant *Pheidole dentata*, an ecological competitor (65), did not significantly differ between treatments (Fig. 6, $\chi^2 = 2.11$, $P = 0.35$). This indicates the ability to discriminate non-nestmates from nestmates, and to respond adaptively by generating the motor outputs of aggression, was not negatively impacted by nutritional stress.

Discussion

Brain Energetic Prioritization. Models of neuroenergetic prioritization ("selfish brain theory") (e.g., ref. 11) predict preferential energy allocation to the brain under physiologically stressful conditions. Nutritionally compromised *T. immigrans* workers maintained their absolute brain metabolic rate and strongly elevated the proportion of energy allocated to the brain, likely aiding in their retention of the sensory processing and motor functions needed for effective aggressive behavior. Body metabolic rate and body mass–specific metabolic rate were significantly lower in nutritionally stressed workers than controls *and* workers from queenright colonies, suggesting that food-deprivation rather than removal from their social environment was responsible for these differences. When food-deprived, worker brains utilized approximately 45% of whole-body energy—nearly twice the proportion used by controls, in association with a significant reduction in body metabolic rate. It is unlikely that this is simply due to the selective survival of resilient workers with high relative brain metabolic rates because that would imply reduced variation in brain metabolic rates of food-deprived workers. Instead, we found the variation of relative brain metabolic rate in food-deprived workers (SD = 18.3) was greater than that of controls (SD = 0.1). Our results demonstrate brain energy prioritization during nutritional stress based on the most direct test to date and suggest that this is likely a general pattern across diverse animals. Brains

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

Table 1. Summary statistics of general linear mixed model results of fixed effects and interactions (indicated by hyphens)

Response Variable	Fixed Effects	Effect Size	Test statistic	*P* value
Brain Mass (mg)	Treatment	0.03	$\chi^2 = 0.87$	0.65
	Environment	0.02	$\chi^2 = 1.50$	0.47
	Body Mass	0.10	$\chi^2 = 13.62$	<0.01
	Treatment–Body Mass	0.05	$\chi^2 = 7.09$	0.03
Body Mass (mg)	Treatment	0.30	F = 30.48	<0.01
	Environment	0.27	F = 2.14	0.16
Relative Brain Mass (%)	Treatment	0.18	$\chi^2 = 33.25$	<0.01
	Environment	0.17	$\chi^2 = 30.63$	<0.01
Relative Brain Metabolic Rate (%)	Treatment	0.16	F = 10.94	<0.01
	Environment	0.04	F = 0.20	0.82
	Body Mass	0.04	F = 5.79	0.02
	Brain Mass	< 0.01	F = 0.62	0.43
	Treatment–Environment	0.06	F = 1.70	0.15
	Treatment–Body Mass	0.02	F = 1.54	0.22
Brain Metabolic Rate (μW)	Treatment	0.02	F = 1.38	0.25
	Environment	0.05	F = 0.25	0.78
	Body Mass	<0.01	F = 0.08	0.77
	Brain Mass	0.02	F = 2.73	0.11
	Treatment–Environment	0.11	F = 3.88	0.01
	Treatment–Body Mass	0.07	F = 4.96	0.01
Brain Mass-specific Metabolic Rate (μW mg^{-1})	Treatment	0.02	F = 1.20	0.31
	Environment	0.30	F = 0.81	0.51
	Body Mass	0.04	F = 4.56	0.04
	Treatment–Environment	0.24	F = 6.31	<0.01
	Treatment–Body Mass	0.11	F = 8.04	<0.01
Body Metabolic Rate (μW)	Treatment	0.16	F = 10.24	<0.01
	Environment	0.01	F = 0.03	0.97
	Body Mass	0.10	F = 12.77	<0.01
	Treatment–Environment	0.04	F = 0.93	0.45
	Treatment–Body Mass	0.02	F = 1.34	0.26
Body Mass–specific Metabolic Rate (μW mg^{-1})	Treatment	0.34	F = 19.21	<0.01
	Environment	0.30	F = 0.96	0.44
Movement Distance (mm)	Treatment	n/a	$\chi^2 = 32.57$	<0.01
	Environment	n/a	$\chi^2 = 8.76$	0.01

Models for each response variable minimized the number of nonsignificant terms without violating model assumptions. F statistics are from Gaussian distributions. χ^2 statistics are from Inverse-Gaussian distributions. Model diagnostics and selection procedures are described in *SI Appendix*, *Supplementary Methods*. *SI Appendix*, Tables S5–S13 provide a full description of the chosen models in contrast with more complex models, including which variables were initially included in each model prior to model simplification. All models included colony identity as a random effect. Nonsignificant terms were included if excluding them violated model assumptions, if the data did not meet assumptions of alternative models, and/or the data could not be transformed to meet assumptions of a parametric distribution. Excluded terms are not shown in Table 2 but are provided in *SI Appendix*, Tables S5–S13.

appear to be disproportionately costly across phylogenetically distant taxa differing in seven orders of magnitude in body size, characterized by divergence in respiratory physiology, and exhibit energy prioritization during nutritional stress.

Maintenance of Behavioral Performance under Energetic Stress. Brain energetic priority appears to preserve defensive capability significant to territory maintenance and thus colony fitness despite a reduction in whole-body metabolic rate. Our results show that worker brains maintain sensory functions, likely the perception and processing of olfactory information in the chemical signatures of cuticular hydrocarbons required for competitor recognition (58–60) and motor functions governing aggression, processes that are under strong neuromodulatory control (66), even during extreme dietary stress. Surprisingly, food deprivation did not decrease worker movement in food-deprived workers in comparison with other workers. Instead, both controls and food-deprived workers reduced movement compared to those from queenright workers. Therefore, our data suggest the reduced movement in food-deprived workers was not due to reduced energy available for movement (controls had access to food, and body metabolic rate did not differ between queenright workers and controls), or for information-processing regulation of motor output (brain metabolic rate did not differ between queenright workers and controls). Rather, our data suggest that the relatively long-term removal of controls and food-deprived workers from colony social context may have suppressed general locomotory activity.

Although we were unable to record worker metabolic rates in vivo during responses to an interspecific competitor, the motor outputs needed for colony defense are known to correlate with reduced brain oxidative phosphorylation (67, 68). Neurophysiological pathways underpinning defensive behavior may overlap with those that initiate nutritional stress challenges. For example, octopamine modulates aggression in some social insects (69, 70), and insulin signaling regulates nutritional and metabolic processes (71, 72). Our results suggest that neurophysiological changes associated with aggression or

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

Table 2. Pairwise statistical comparisons between treatments, defined as Q–C (Queenright–Controls), Q–FD (Queenright–Food-deprived), and C–FD (Controls–Food-deprived)

Response Variable	Contrast	Estimate	SE	Test statistic	*P* value
Brain Mass (mg)	Q–C	0.027	0.052	Z = 0.53	0.86
	Q–FD	0.023	0.044	Z = 0.51	0.87
	C–FD	0.050	0.054	Z = 0.93	0.62
Body Mass (mg)	Q–C	0.257	0.046	T = 5.66	<0.01
	Q–FD	0.022	0.043	T = 0.50	0.87
	C–FD	0.279	0.038	T = 7.43	<0.01
Relative Brain Mass (%)	Q–C	0.011	0.002	Z = 4.80	<0.01
	Q–FD	0.002	0.003	Z = 0.61	0.81
	C–FD	0.013	0.032	Z = 4.72	<0.01
Relative Brain Metabolic Rate (%)	Q–C	0.020	0.314	T = 0.64	0.80
	Q–FD	0.104	0.029	T = 3.55	<0.01
	C–FD	0.124	0.029	T = 4.26	<0.01
Brain Metabolic Rate (μW)	Q–C	0.043	0.045	T = 0.94	0.61
	Q–FD	0.064	0.042	T = 1.53	0.28
	C–FD	0.021	0.040	T = 0.53	0.86
Brain Mass-specific Metabolic Rate (μW mg^{-1})	Q–C	0.043	0.071	T = 0.60	0.82
	Q–FD	0.103	0.067	T = 1.55	0.27
	C–FD	0.061	0.073	T = 0.83	0.69
Body Metabolic Rate (μW)	Q–C	0.018	0.089	T = 0.21	0.98
	Q–FD	0.321	0.083	T = 3.88	<0.01
	C–FD	0.339	0.088	T = 3.86	<0.01
Body Mass–specific Metabolic Rate (μW mg^{-1})	Q–C	0.145	0.076	T = 1.92	0.14
	Q–FD	0.433	0.073	T = 5.98	<0.01
	C–FD	0.288	0.070	T = 4.11	<0.01
Movement Distance (mm)	Q–C	0.707	0.141	Z = 5.01	<0.01
	Q–FD	0.657	0.149	Z = 4.42	<0.01
	C–FD	0.050	0.162	Z = 0.31	0.95

responses to food deprivation differ, as worker brain metabolic rate was not significantly reduced during food deprivation. Defensive behavior mandates a rapid allocation of energy investment associated with an extremely high risk of mortality whereas starvation resistance requires physiological plasticity for long-term balance. Future studies may identify conserved mechanisms that regulate physiological responses to social challenge (73).

Mechanisms of Brain Energetic Prioritization. Insects and vertebrates may share mechanisms for maintaining brain metabolism under physiological stress. Models of brain energetic priority have suggested that human brains utilize neuroendocrine mechanisms to reduce glucose uptake by peripheral organs to save fuel for brain metabolic demands (74). In response to reduced cerebral intracellular ATP, glutamate release through the sympathoadrenal system inhibits pancreatic insulin secretion, reducing glucose uptake by muscle and adipose tissue and increasing glucose availability to the brain (75). Insects may have a similar mechanism of neuroendocrine regulation to ensure the energy needs of the brain are met. Trehalose, the circulating carbohydrate of insects, is stored in the fat body as glycogen (71, 76) in response to insulin-like protein (ILP) signaling (77, 78). When hemolymph trehalose concentration is low, adipokinetic hormone (79–82) and inhibition of ILP secretion (83, 84) increase hemolymph carbohydrate levels, suggesting neuroendocrine regulation of energy demands (85). During starvation, glycogen and triglyceride levels in the honeybee brain increase in association with a decrease in carbohydrate levels in the thorax and fat body (86). Therefore, although insulin signaling has evolved in social insects as a regulator of phenotypic plasticity and division of labor (72, 87–89), a more ancient mechanism in which insulin suppression mobilizes energy reserves to increase allocation to the brain may be conserved across vertebrates and insects, consistent with the deep homology between invertebrate and vertebrate brains (57). Mammalian and insect glia have metabolic pathways controlling the synthesis and mobilization of lipids to neurons during glucose depletion (90), further supporting the conservation of mechanisms to maintain expensive brain energetic demands during metabolic stress and common metabolic scaling.

Measurement of Ex-Vivo Brain Metabolic Rate. Although excision eliminates synaptic inputs to the brain, we are confident our ex vivo measurements reflect treatment effects in vivo. Axotomy of a small number of neurons does not necessarily cause significant damage to alter brain energy demands; it results in rapid depolarization of cut neurons due to sodium and calcium ion influx, causing a spike in energy use for repolarization. If ion pumps are overwhelmed, apoptosis may result, although neurons can mechanistically prevent apoptosis in response to axotomy (91). However, it is clear in our study that the number of sensory and motor neurons involved in brain excision is a minute fraction of the total number of brain cells (roughly 40,000 to 70,000 in size-equivalent ant species (92). Therefore, it is unlikely that any potential increase in energy demands of several axotomized neurons, or their apoptosis, would cause physiological damage to the brain and affect the rates we measured. Our recordings clearly demonstrate that brain metabolism remains stable for several hours (*SI Appendix*, Fig. S12). Axotomy of a few neurons therefore does not increase metabolic rate or cause apoptosis throughout the brain.

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

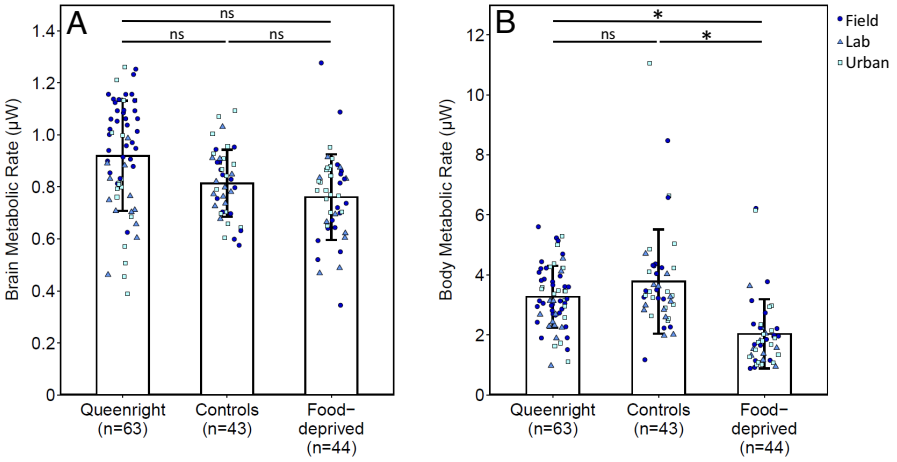


Fig. 2. (*A*) Worker brain metabolic rate was not significantly affected by treatment (Table 1, F = 1.38, *P* = 0.25) or colony-collection environment (Table 1, F = 0.25, *P* = 0.78). Bar plots represent means ± SD. (*B*) Worker body metabolic rate was significantly affected by treatment (Table 2, F = 10.24, *P* < 0.01), but not colony-collection environment Table 2, F = 0.03, *P* = 0.97). Food-deprived workers exhibited significantly lower body metabolic rates than controls (Table 2, T = 3.86, *P* < 0.01) and queenright workers (Table 2, T = 3.88, *P* < 0.01). Bar plots represent means ± SD.

Previous studies demonstrate that although in vivo brain metabolism is usually higher than ex vivo metabolism, the magnitude of variation and the effect of treatment as determined from in-vivo and ex-vivo measurements are consistent. Buschiazzo et al. (16) show that ex vivo rat brain glucose consumption rates of food-deprived and control groups are both approximately 50% of in vivo brain glucose consumption rates and Oikawa and Itazawa (93, 94) found that ex vivo tissue metabolic rates sum to approximately one third of whole-body metabolic rates (see *SI Appendix, Supplementary Methods* for further details). Therefore, if our ex vivo brain metabolic rates differ from in vivo metabolic rates as a result of eliminating synaptic inputs, previous studies suggest that our measurements underestimate the fraction of whole-body metabolic rate allocated to brain metabolism, strongly supporting the primary conclusions of our study and suggesting that our inference that worker relative brain metabolic rate is disproportionately high in comparison to worker relative brain mass is conservative.

Conservation of Neurometabolic Scaling across Phylogenetically Distant Taxa. In support of the hypothesis that a common pattern of brain energy prioritization result from conserved brain processes driving high metabolic expenses, we found that the average brain metabolic rate of *T. immigrans* queenright workers is within the CI of the regression of mammalian brain metabolic rate on brain mass (Fig. 7). This hypometric pattern is common across *T. immigrans* and mammalian brains (log–log slope of 0.87) and suggests conserved metabolic scaling. Neurometabolism appears to be expensive across all species measured to date (23, the present study). Therefore, despite differences in phylogenetic history, brain size and physiology, and the sociobiology of ants and mammals, our results suggest that the high costs of the "expensive brain" are evolutionarily ancient and are required to maintain membrane potentials and cellular communication (50, 95–97). The role of collective intelligence on brain size evolution in ants and humans (43, 44) may in part explain this neurophysiological similarity.

The hypometric slope of brain metabolic rate on brain mass shared by ants and mammals suggests similar physiological interactions between body size and brain function. This body size effect is illustrated by the approximately 3× higher brain mass-specific metabolic rates in ants (~0.102 mL min^{-1} g^{-1} in *T. immigrans* queenright workers, corrected to 37 °C assuming Q_{10} = 2) compared to humans (~0.035 mL min^{-1} g^{-1}) (24). Factors contributing to the evolution of hypometric scaling of brain metabolic rate are unclear. Smaller-bodied animals may be under stronger selection for mass-specific neuro-locomotory performance (98), thus driving higher relative brain size and metabolic rate to maintain behavioral performance. Smaller animals also tend to have relatively larger sensory structures, which may require relatively larger brains with higher mass-specific energy needs for processing sensory information (98–102). Smaller-bodied animals may also be more

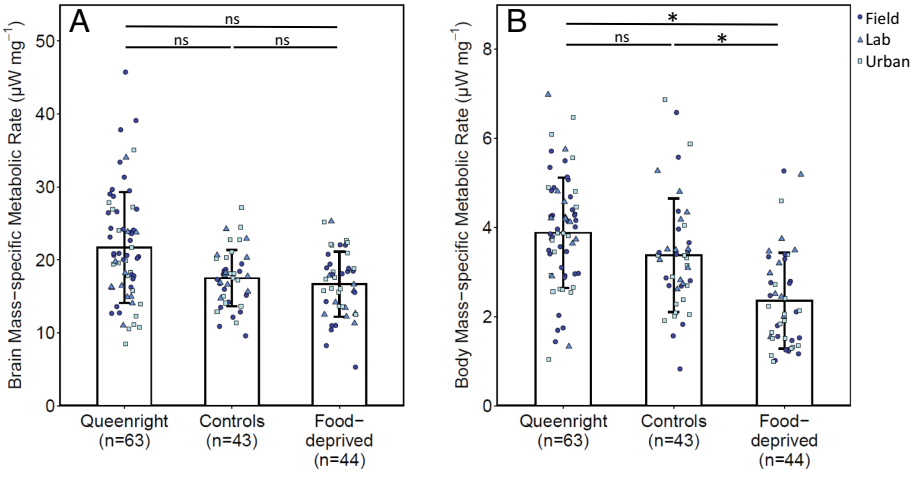


Fig. 3. (*A*) Brain mass-specific metabolic rate was not significantly affected by treatment (Table 1, F = 1.20, *P* = 0.31) or colony-collection environment (Table 1, F = 0.81, *P* = 0.51), but these did have a significant interaction effect (Table 1, F = 6.31, *P* < 0.01). Bar plots represent means ± SD. (*B*) Treatment had a significant effect on body mass–specific metabolic rate (body metabolic rate divided by body mass) (Table 1, F = 19.21, *P* < 0.01). The colony-collection environment did not have a significant effect (Table 1, F = 0.96, *P* = 0.44). Food-deprived workers exhibited significantly lower body mass–specific metabolic rates than controls (Table 2, T = 4.11, *P* < 0.01) and queenright workers (Table 3, T = 5.98, *P* < 0.01). Bar plots represent means ± SD.

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

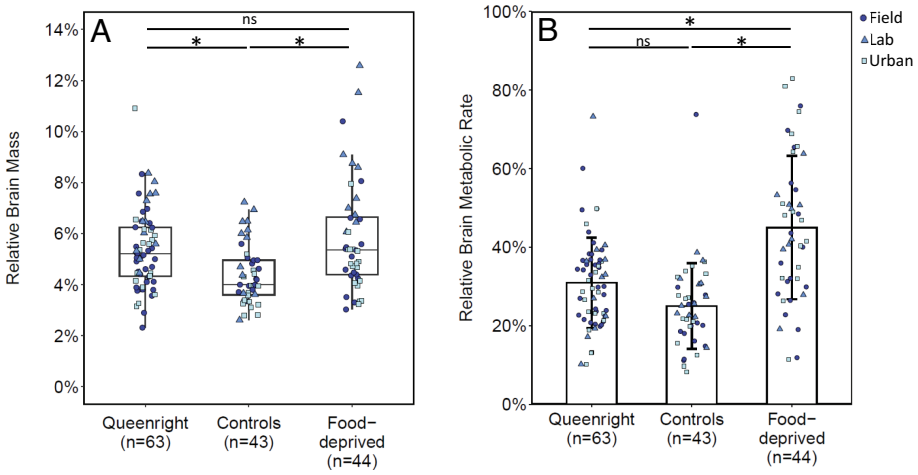


Fig. 4. (*A*) Worker relative brain mass (brain mass as a proportion of body mass) was affected by treatment (Table 1, $\chi^2 = 33.25$, $P < 0.01$) and colony environment (Table 1, $\chi^2 = 30.63$, $P < 0.01$). Controls had significantly lower relative brain mass than queenright workers (Table 2, Z = 4.80, $P < 0.01$) and food-deprived workers (Table 3, Z = 4.72, $P < 0.01$). Box plots represent medians ± 150% of the interquartile range. (*B*) Worker relative brain metabolic rates (brain metabolic rate as a proportion of body metabolic rate) were significantly affected by treatment (Table 1, F = 10.94, $P < 0.01$) but not colony environment (Table 1, F = 0.20, $P = 0.82$). Food-deprived workers exhibited significantly higher relative brain metabolic rates than controls (Table 2, T = 4.26, $P < 0.01$) and queenright workers [Table 2, T = 3.55, $P < 0.01$)]. Bar plots represent means ± SD.

strongly selected for higher growth rates and associated foraging skills, selecting for the evolution of relatively larger and more metabolically active brains (103, 104). Independent of the evolutionary processes involved, our study is to our knowledge the first to demonstrate that insects share a common hypometric brain metabolic scaling pattern with mammals. Examination of the scaling of brain metabolism within and across ants and other insects is necessary to identify the selective forces and physiological mechanisms regulating the hypometric scaling of both brain and body metabolic rate across diverse species.

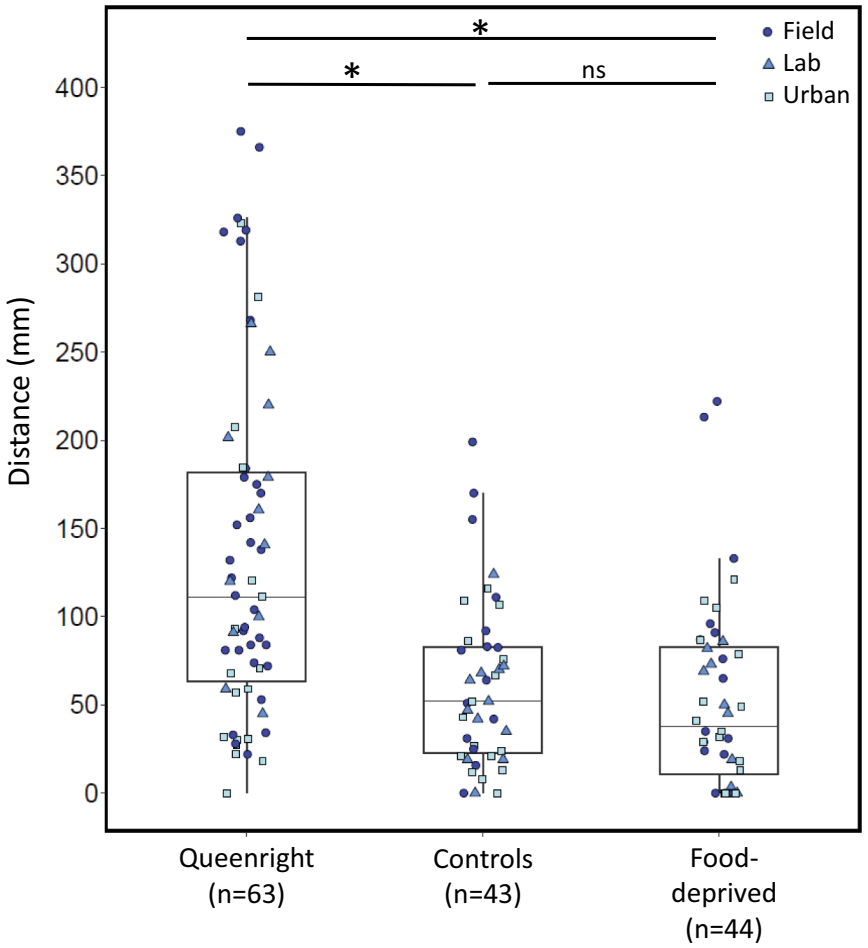


Fig. 5. Treatment had a significant effect on worker movement (Table 1, $\chi^2 = 32.57$, $P < 0.01$). Queenright workers moved significantly more than controls (Table 2, Z = 5.01, $P < 0.01$) and food-deprived workers (Table 2, Z = 4.42, $P < 0.01$). The environment also had a significant effect on worker movement (Table 2, $\chi^2 = 8.76$, $P = 0.01$). Box plots represent medians ± 150% of the interquartile range.

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

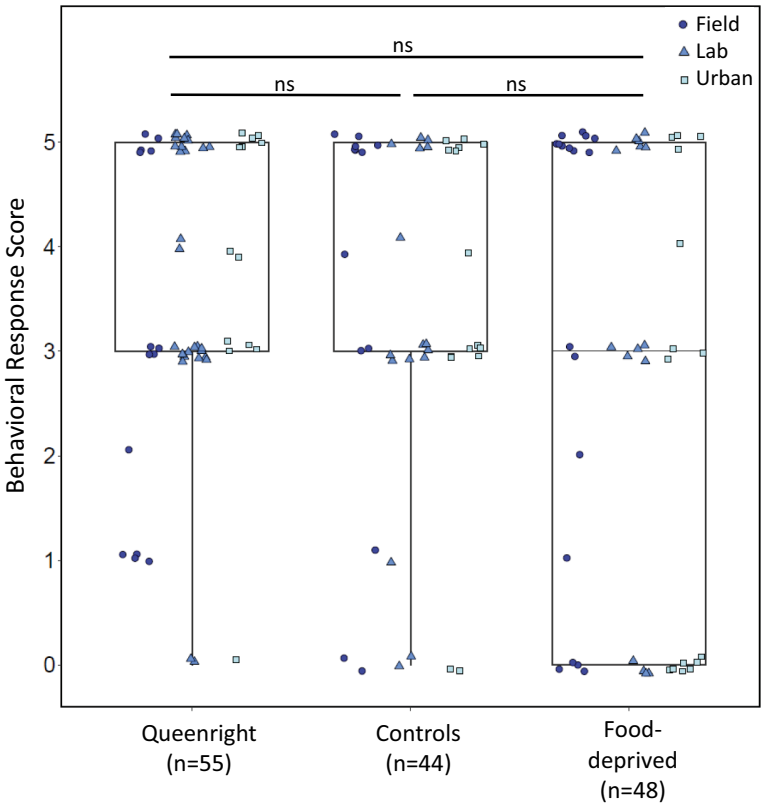


Fig. 6. Worker behavioral responses were not significantly impacted by treatment (Kruskal–Wallis: χ^2 = 2.11, P = 0.35) or colony environment (Kruskal–Wallis: χ^2 = 0.35, P = 0.84). Box plots represent medians ± 150% of the interquartile range. Points are staggered vertically and horizontally to reduce visual overlap.

Hypometric brain metabolic scaling in eusocial insects is also likely influenced by the complexity of social life and worker behavioral demands (40, 52, 55). Workers of the highly complex Australasian weaver ant *Oecophylla smaragdina* have larger absolute mushroom body volume, the brain compartment responsible for higher-order multimodal information processing, but lower mushroom body cytochrome oxidase activity, a proxy for neuron metabolic rate, than the socially basic ant *Formica subsericea* (45). This suggests that increased brain investment may be offset by lower brain operation costs associated with variation in social complexity. Differences in mushroom body metabolic costs may be due to variation in sparse coding, an energy efficient mechanism of sensory encoding (4) exhibited by the mushroom bodies (105, 106). Our study demonstrates that *T. immigrans* worker brains

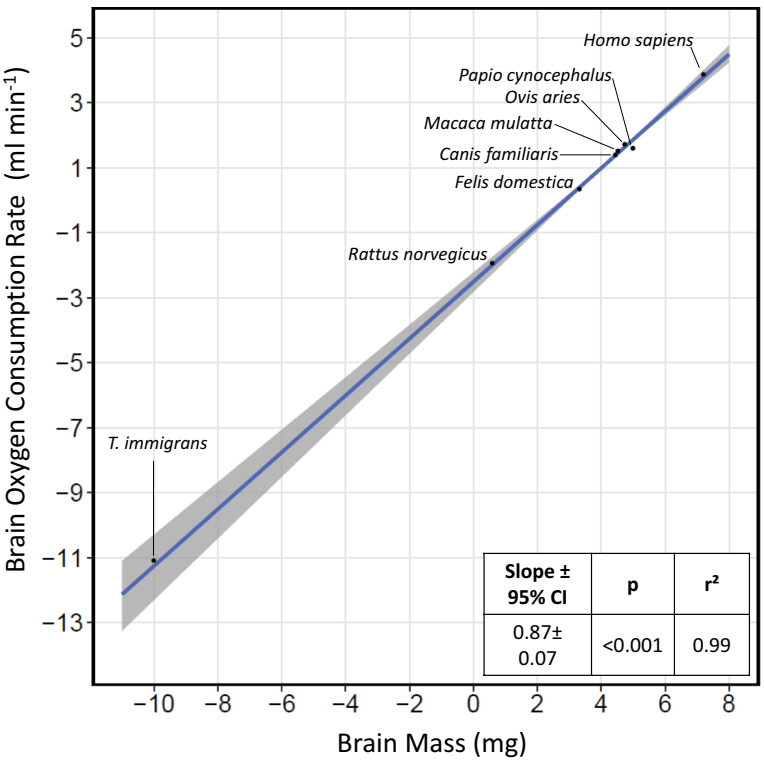


Fig. 7. Brain metabolic scaling of *T. immigrans* and mammals for which data is available. Mammalian brain OCR scales hypometrically with a slope of less than 1 and *T. immigrans* worker brain OCR falls within the CI of the mammalian regression. Brain OCRs for mammals were calculated as averages of minimally two studies with the exception of *Papio cynocephalus* (one study). Mammalian brain OCR and brain mass data are from Karbowski (24), Mink et al. (23), and references therein. *T. immigrans* brain OCR was recorded at 31 °C. Assuming a Q_{10} of 2, the average brain OCR of *T. immigrans* queenright workers (9.54×10^{-6} mL O_2 min^{-1}) was corrected to 1.53×10^{-5} mL O_2 min^{-1} to approximate worker brain OCR at 37 °C, the temperature at which brain OCR of most mammalian species was recorded.

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

have disproportionately high metabolic rates relative to other tissues. However, a socially complex species with similar worker body size may have larger brain size, but lower brain metabolic rates. Comparisons of brain metabolic rates of *T. immigrans* with other species of equivalent or variable body size and species that vary in social complexity are needed to assess whether worker brain metabolic scaling is influenced by sociobiological variation.

Conclusions

Our results provide a direct assay that confirms energy is preferentially allocated to the brain during nutritional stress, suggesting that neurophysiological prioritization is broadly conserved across phylogenetically diverse clades. *T. immigrans* workers, like humans, have energetically expensive brains: Workers allocate 24 to 30% of their body metabolism to neurometabolism although representing ~5% of worker body mass. Energy allocation nearly doubled under conditions of food deprivation due to a significant reduction in body metabolic rate in the absence of a comparable reduction in brain metabolic rate. The mechanisms underlying energy prioritization to meet brain metabolic demands may be conserved or functionally convergent, impacted by selection to maintain information-processing and motor systems that maintain critical behaviors under conditions of potentially lethal physiological stress.

Materials and Methods

Study Species, Colony Collection, and Laboratory Culture. The ant *T. immigrans* (=*T. caespitum*) is globally distributed and adapted to fields, forests, and urban environments. This species has an opportunistic generalist and seasonally variable diet (107) and feeds on human foods (108) in cities. Social insect diets affect worker metabolic rates (109, 110) and fitness (62). Therefore, we sampled *T.* workers from different colony collection environments to ensure broad representation of dietary variation and its potential metabolic influences. Sampling included 8 colonies located at the Middlesex Fells Reservation, Stoneham, MA, and Cleveland Circle, Boston, MA (field colonies), 8 colonies located at Boston University (urban colonies), and 2 colonies reared from newly inseminated queens collected at Newburgh, NY, and Boston, MA (lab colonies). Laboratory colony husbandry is described in detail in *SI Appendix, Supplementary Methods*. Colonies were collected in compliance with the containment conditions of USDA APHIS Permit PPQ526P–12–04067. We collected sufficient samples for study while minimizing impact on colony health.

To control for age-related bias in sampling, we exclusively sampled fully pigmented individuals foraging outside of their nest and thus likely to be older individuals. Additionally, our sampling included a total of 2,760 workers distributed among 92 subcolonies, each composed of 30 workers. Workers were collected randomly from 18 source colonies that may have differed in size and age demography, potentially affecting the average age of foraging workers. However, these several thousand foraging workers were randomly distributed between controls and food-deprived subcolonies, likely distributing workers of varying age and resilience equally among all treatments.

Nutritional Stress. Food deprivation can decrease body metabolic rate in ants (53) and have pervasive effects on development in honeybees (86), but the effect of dietary stress on brain metabolism has not been studied in insects. To determine the effect of food deprivation on brain and whole-body metabolism, queenless subcolonies, each consisting of groups of 30 workers randomly collected from lab, field, and urban colonies were provided either a lab colony diet (referred to as controls) or only water-soaked cotton for hydration (referred to as food-deprived workers) on alternate days for 14 to 40 d. All subcolonies were maintained in 15 × 8 × 4 cm plastic boxes, each containing a 3 cm diameter Petri dish lined with hydrated dental stone and covered in red cellophane. Sampling was paired for food-deprived workers and controls such that an equal number of workers from a control subcolony and food-deprived subcolony exposed to the same treatment duration were sampled when the food-deprived subcolony had 80% worker mortality (5 to 6 workers remaining). To examine the effect of long-term removal from the social context of a queenright colony, body metabolic rates were also recorded within several hours of worker removal from queenright colonies to allow us to distinguish the influence of worker diet manipulation from the effect of social separation from the parent colony on worker metabolism.

To confirm that food-deprivation caused significant physiological stress, we compared survival probability curves of controls and food-deprived workers (*SI Appendix*, Fig. S11) using a Weibull AFT regression to assess variation in survival across treatments and environments. Survival probability was significantly lower for food-deprived workers relative to controls (hazard = −1.58 ± 0.06, Z = 26.12, $P < 0.001$). Additional statistics of survival probability are provided in *SI Appendix, Supplementary Results*. These data indicate that food-deprived workers were significantly physiologically stressed at the time of sampling. Workers that were alive, but moribund, were not included in metabolic assays.

Brain Excision and Measurement of Brain Metabolism. Workers were weighed on a Mettler AE163 analytical balance. To prevent damage to worker brains, heads of decapitated workers were immediately placed in Seahorse media and head capsules were cut longitudinally beneath the mandibles, allowing the ventral surface of the head to be removed to expose the brain and attached nerves. Nerves originating from the antennal lobe and subesophageal zone were carefully cut without damaging these brain compartments. After these nerves were cut, the brain was carefully excised from the head capsule. Excised brains were immediately placed in Seahorse XF base medium supplemented with 10 mM sodium pyruvate and 10 mM glucose corrected to pH 7.0 and placed into Seahorse XF HS mini cartridges. At no point were brains removed from the media during this process. Intact, undamaged brains were secured in Seahorse plate wells by custom-made Delrin screens to minimize brain movement during measurement (29). We only included data from excised brains that were undamaged and maintained approximately constant metabolic rates during measurement. A representative image of an excised, undamaged brain is included in *SI Appendix*, Fig. S17.

Metabolic data were recorded from worker brains ex vivo using an Agilent Seahorse XF-HS Mini Analyzer within 2.5 h of brain excision. Oxygen consumption rate (OCR) was recorded at 31 ± 0.03 °C, consistent with the thermal preference of *T. immigrans* (111). All OCR data were converted to units of μWatts for comparison with body metabolic rates. Details of brain metabolic data recording and analysis are provided in our *SI Appendix, Supplementary Methods*. Representative traces of stable brain OCR recordings are presented in *SI Appendix*, Fig. S12. Following brain OCR recordings, brains were transferred to tared foil using Dumont-5 forceps, wicked dry using a Kimwipe and weighed on a Sartorius CP2P 191 microbalance. Additional details of brain metabolism measurement are provided in *SI Appendix, Supplementary Methods* section on Measurement of Brain Metabolism.

Measurement of Body Metabolism. Our goal was to measure the whole-body metabolic rate inclusive of effects of nutritional stress on behavior. We therefore used closed system respirometry to assess CO_2 emission rates of ants for 60 to 90 min, during which workers could exhibit behaviors such as walking. Static-Injection respirometry was performed using a Sable Systems Fox Box to record CO_2 emission from individual workers (methods adapted from Lighton) (112) at 31 ± 0.19 °C. Analysis of CO_2 emission data was performed in Expedata which recorded $\%CO_2$, temperature, and flow rate on separate channels according to Lighton (112). All CO_2 emission rates were converted to units of μWatts for comparison with brain metabolic rates. Details on the respirometry setup and analysis of CO_2 emission data are provided in our *SI Appendix, Supplementary Methods*.

Assays of Effects of Nutritional Stress on Worker Locomotion and Anti-competitor Behaviors. To determine the effect of food deprivation on general motor output, worker movement was recorded during the closed system respirometry using a Vixia HF R500 video camera. Worker path length was estimated by the number of times a worker's mesosoma crossed markers of distance traveled during three 1-min periods (0 to 1 min, 30 to 31 min, and 59 to 60 min) within the respirometry chamber. Only movement parallel or perpendicular to the longitudinal axis of the syringe was used for analysis; diagonal movement was ignored. Movement was recorded for all workers sampled for body metabolic rates. Recordings of 11 control workers were excluded from analyses of movement, mass and metabolism because videos were incomplete and data were not comparable with recordings of other workers.

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

Neurometabolism supports sensory information processing demands and adaptive behavior, such as the discrimination of nestmates and nonnestmates and generation of motor outputs underpinning aggression. To assess recognition and responsiveness to contact with a competitive ant species as well as defensive motor output, we staged encounters between one *T. immigrans* worker and one *P. dentata* minor worker. *P. dentata* is sympatric with *T. immigrans* in the Southeastern United States of America (113) and *T. immigrans* elicits an alarm-defense response in *P. dentata* colonies (65, 114), indicating a competitive ecological relationship. *P. dentata* thus serves as a natural competitor expected to be immediately recognized by workers of *T. immigrans*, leading to an aggressive response. Our assay allows us to determine if the components of nestmate recognition–olfactory detection of cuticular hydrocarbon profiles, higher-order integration and processing, and the genesis of the motor responses of aggression–are compromised by lower availability of energy necessary to sustain these functions by the brain. A *T. immigrans* control worker, food-deprived worker, or worker from a queenright colony, was placed in a Fluon®-lined Petri dish (8 cm diameter). One minute was allowed for acclimation to the assay arena, followed by the introduction of a *P. dentata* minor worker opponent. Ambient temperature was maintained at approximately 31 °C. Behaviors relevant to the recognition of a worker of a competing species and resulting agonistic actions were recorded on a scale of progressively increasing levels of aggression: 1) No Response: brief contact between workers without interaction; 2) Olfactory Assessment: antennae extended toward the opponent; 3) Mandibular Flaring: opening of the mandibles following contact; 4) Attack posture: gaster oriented anteriorly beneath the mesosoma in preparation to sting; and 5) Biting: attacking with mandibles. Focal *T. immigrans* worker behavior was recorded for 3 min or until biting occurred, in which case workers were separated to prevent mortality. If workers engaged in multiple interactions during the assay (e.g., mandible flaring and biting), the highest score across all interactions was used for data analysis. When possible, workers different from those used to record body metabolic rate were assayed. However, if workers were assayed to assess responsiveness and body metabolic rate, they were isolated for at least 3 h in a moist, dental stone-lined Petri dish (4 cm diameter) prior to recording body metabolic rate recording to avoid impacts of aggression on metabolism (e.g., ref. 67).

Statistical Analysis. All statistics were performed in RStudio (v. 2024.04.2) and a critical $\alpha = 0.05$ was used to assess all statistical differences. A Kruskal–Wallis Chi-squared test was used to assess statistical differences in behavioral response scores between treatments. For all other response variables, we used a general linear mixed model through the *lmer* package (115) to assess statistical differences with treatment (Queenright, Controls, Food-deprived) and colony collection environment (Field, Lab, Urban) as fixed effects and colony identity (the source colony from which a worker was sampled) as a random effect. Details on model diagnostics, additional included fixed effects, and our model selection procedure are provided in Table 1 and *SI Appendix*, *Supplementary Methods*, and results of this procedure for each response variable are provided in *SI Appendix*, Tables S5–S12. All figures were generated in RStudio using the *ggplot2* package (116). Bar plots of means were used to represent parametric data; box plots of medians were used to represent nonparametric data. The *position_jitterdodge* function was applied to reduce visual overlap of points without changing measurement values.

Metabolic rate scales with size (54). Therefore, metabolic and mass scaling analyses were performed using standard major axis regression using the *SMATR* package in R (117) to inform whether to include these variables as fixed effects in our models. Due to significant scaling of log brain mass on log body mass, log body metabolic rate on log body mass, and log brain metabolic rate on log body metabolic rate in at least one treatment, body mass was included as an initial fixed effect in our models of brain mass, body metabolic rate, brain metabolic rate, brain mass-specific metabolic rate, and relative brain metabolic rate, while body metabolic rate was included as an initial fixed effect for models of brain metabolic rate and brain mass-specific metabolic rate. There was no environment effect on brain mass and metabolic rate or body mass and metabolic rate and therefore interactions between colony-collection environments and these models were not included in any model. Variables were sequentially dropped from models if they did not have significant effects on the response variable, starting with those insignificant effects with the lowest effect sizes. Scaling statistics are reported in *SI Appendix*, Figs. S13–S16 and *Supplementary Results*.

We report data on absolute rates of body and brain metabolism as well as mass-specific metabolic rates (body metabolic rate/body mass; brain metabolic rate/brain mass). To examine the effect of food deprivation on the proportion of mass and energy allocated to the brain, we also report relative brain mass (brain mass/body mass) and relative brain metabolic rate (brain metabolic rate/body metabolic rate). Workers that were moribund or dead at the time of excision were excluded from analysis. Only intact, nondamaged brains were used for brain metabolism and brain mass measurement. A representative image of a nondamaged brain is provided in *SI Appendix*, Fig. S17.

Statistics of general linear mixed model results for main and interaction effects are summarized in Table 1. Statistical results of pairwise comparisons between treatments pooled across environments of colony collection are summarized in Table 2. Data differentiated by treatment and environment of colony collection are presented in *SI Appendix*, Figs. S1–S10. Significant treatment–environment effects were only found for brain metabolic rate and brain mass-specific metabolic rate and comparisons between treatments within each environment of colony collection are provided in *SI Appendix*, Tables S2 and S3. Pairwise comparisons were performed on estimated marginal means using the *emmeans* package in R (118). Data modeled using a Gaussian distribution are presented as bar plots of means ± SD. Data modeled using an inverse Gaussian distribution are presented as box plots of medians and the interquartile range.

Data, Materials, and Software Availability. All study data are included as supplementary material in Dataset S1. A README file for Dataset S1 is included as Dataset S2. The R code used in statistical analysis is provided as Dataset S3.

ACKNOWLEDGMENTS. We thank Dr. Marla Tipping for introducing us to the Agilent Seahorse XF Analyzer and Dr. Jane Waters for contributing to early discussions of metabolic measurements and technical advice using the Sable Systems FoxBox. We thank Dr. Pamela Templer for use of the Sartorius CP2P 191 microbalance. Jordan Smith and Dr. Frank Azorsa provided constructive comments during manuscript development. This research was supported by NSF grant IOS 1953393 to J.F.A.T., NFS IOS 1953419 to J.F.H. and by grant CNS2023-143722 funded by MCIN/AEI/ 10.13039/501100011033 and by European Union NextGenerationEU/PRTR to S.A. During the duration of this research, S.A. was additionally supported by a Marie Sklodowska-Curie Individual Fellowship BrainiAnts-660976 and Ayudas destinadas a la atracción de talento investigador a la Comunidad de Madrid en centros de I+D.

Author affiliations: [a]Department of Biology, Boston University, Boston, MA 02215; [b]Department of Biology, Stanford University, Stanford, CA 94305; [c]Departamento de Biología, Universidad Rey Juan Carlos, Móstoles 28933, Spain; [d]Instituto de Investigación en Cambio Global, Universidad Rey Juan Carlos, Móstoles 28933, Spain; and [e]School of Life Sciences, Arizona State University, Tempe, AZ 85287-4501

1. J. Brožek, F. Grande, Body composition and basal metabolism in man: Correlation analysis versus physiological approach. *J. Hum. Biol.* **27**, 22–31 (1955).
2. S. S. Kety, Human cerebral blood flow and oxygen consumption as related to aging. *J. Chronic. Dis.* **3**, 478–486 (1956).
3. M. Elia, "Organ and tissue contribution to metabolic rate" in *Energy Metabolism: Tissue Determinants and Cellular Corollaries*, J. M. Kinney, H. N. Tucker, Eds. (Raven Press, 1992), pp. 61–79.
4. D. Attwell, S. B. Laughlin, An energy budget for signaling in the grey matter of the brain. *J. Cereb. Blood Flow Metab.* **21**, 1133–1145 (2001).
5. L. C. Aiello, P. Wheeler, The expensive-tissue hypothesis: The brain and the digestive system in human and primate evolution. *Curr. Anthropol.* **36**, 199–221 (1995).
6. K. Isler, C. P. van Schaik, The expensive brain: A framework for explaining evolutionary changes in brain size. *J. Hum. Evol.* **57**, 392–400 (2009).
7. A. Navarrete, C. P. van Schaik, K. Isler, Energetics and the evolution of human brain size. *Nature* **480**, 91–93 (2011).
8. H. Pontzer *et al.*, Metabolic acceleration and the evolution of human brain size and life history. *Nature* **533**, 390–392 (2016).
9. R. Wrangham, Control of fire in the Paleolithic: Evaluating the cooking hypothesis. *Curr. Anthropol.* **58**, S303–S313 (2017).
10. S. A. Heldstab, K. Isler, S. M. Graber, C. Schuppli, C. P. van Schaik, The economics of brain size evolution in vertebrates. *Curr. Biol.* **32**, R697–R708 (2022).
11. A. Peters, The selfish brain: Competition for energy resources. *Am. J. Hum. Biol.* **23**, 29–34 (2011).

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

12. A. Peters, B. Kubera, C. Hubold, D. Langemann, The corpulent phenotype–How the brain maximizes survival in stressful environments. *Front. Neurosci.* **7**, 47 (2013).
13. Z. Padamsey, N. L. Rochefort, Paying the brain's energy bill. *Curr. Opin. Neurobiol.* **78**, 102668 (2023).
14. O. E. Owen *et al.*, Brain metabolism during fasting. *J. Clin. Invest.* **46**, 1589–1593 (1967).
15. K. M. Oltmanns *et al.*, Differential energetic response of brain vs skeletal muscle upon glycemic variations in healthy humans. *Am. J. Physiol. Reg. Integr. Comp. Physiol.* **294**, R12–R16 (2008).
16. A. Buschiazzo *et al.*, Effect of starvation on brain glucose metabolism and 18F-2-fluoro-2-deoxyglucose uptake: An experimental in-vivo and ex-vivo study. *EJNMMI Res.* **8**, 44 (2018).
17. M. Sprengell, B. Kubera, A. Peters, Brain more resistant to energy restriction than body: A systematic review. *Front. Neurosci.* **15**, 63917 (2021).
18. M. Sprengell, B. Kubera, A. Peters, Proximal disruption of brain energy supply raises systemic blood glucose: A systematic review. *Front. Neurosci.* **15**, 685031 (2021).
19. M. Sprengell, B. Kubera, A. Peters, Brain mass (energy) resistant to hyperglycaemic oversupply: A systematic review. *Front. Neurosci.* **15**, 740502 (2021).
20. B. Stapel *et al.*, In vivo magnetic resonance spectrometry imaging demonstrates comparable adaptation of brain energy metabolism to metabolic stress induced by 72h of fasting in depressed patients and healthy volunteers. *J. Psychiatr. Res.* **143**, 422–428 (2021).
21. J. T. Grist *et al.*, Quantifying normal human brain metabolism using hyperpolarized [1-^{13}C] pyruvate and magnetic resonance imaging. *Neuroimage* **189**, 171–179 (2019).
22. S. N. Vandekar *et al.*, Sex differences in estimated brain metabolism in relation to body growth through adolescence. *J. Cereb. Blood Flow Metab.* **39**, 524–535 (2019).
23. J. W. Mink, R. J. Blumenschine, D. B. Adams, Ratio of central nervous system to body metabolism in vertebrates: Its constancy and functional basis. *Am. J. Physiol.* **241**, R203–R212 (1981).
24. J. Karbowski, Scaling of brain metabolism and blood flow in relation to capillary and neural scaling. *PLoS One* **6**, e26709 (2011).
25. K. von Eugen *et al.*, Avian neurons consume three times less glucose than mammalian neurons. *Curr. Biol.* **32**, 4306–4313 (2022).
26. G. Wegener, Brains burning fat: Different forms of energy metabolism in the CNS of insects. *Naturwissenchaften* **70**, 43–45 (1983).
27. G. Wegener, "Insect brain metabolism under normoxic and hypoxic conditions" in *Arthropod Brain: Its Evolution, Structure, Development and Function*, A. P. Gupta, Ed. (John Wiley & Sons, 1987), pp. 369–397.
28. M. J. Kern, Metabolic rate of the insect brain in relation to body size and phylogeny. *Comp. Biochem. Physiol. A* **81**, 501–506 (1985).
29. K. E. Neville *et al.*, A novel ex-vivo method for measuring whole brain metabolism in model systems. *J. Neurosci. Methods* **296**, 32–43 (2018).
30. C. C. Rittschof, S. Schirmeier, Insect models of central nervous system energy metabolism and its links to behavior. *Glia* **66**, 1160–1175 (2018).
31. E. de Tredern *et al.*, Glial glucose fuels the neuronal pentose phosphate pathway for long-term memory. *Cell Rep.* **36**, 109620 (2021).
32. C. Tait, A. J. Chicco, D. Naug, Brain energy metabolism as an underlying basis of slow and fast cognitive phenotypes in honeybees. *J. Exp. Biol.* **227**, jeb247835 (2024).
33. F. Mery, T. J. Kawecki, A cost of long-term memory in Drosophila. *Science* **308**, 1148 (2005).
34. J. M. S. Burger, M. Kolss, J. Pont, T. J. Kawecki, Learning ability and longevity: A symmetrical evolutionary trade-off in *Drosophila*. *Evolution* **62**, 1294–1304 (2008).
35. E. C. Snell-Rood, G. Davidowitz, D. R. Papaj, Reproductive tradeoffs of learning in a butterfly. *Behav. Ecol.* **22**, 291–302 (2011).
36. A. Kotrschal *et al.*, Artificial selection on relative brain size in the guppy reveals costs and benefits of evolving a larger brain. *Curr. Biol.* **23**, 168–171 (2013).
37. I. C. Christiansen, S. Szin, P. Schausberger, Benefit-cost trade-offs of early learning in foraging predatory mites Amblyseius swirskii. *Sci. Rep.* **6**, 23571 (2016).
38. D. Longman, J. T. Stock, J. C. K. Wells, A trade-off between cognitive and physical performance, with relative preservation of brain function. *Sci. Rep.* **7**, 13709 (2017).
39. E. Szathmáry, J. M. Smith, The major evolutionary transitions. *Nature* **374**, 227–232 (1995).
40. J. F. A. Traniello, T. A. Linksvayer, Z. N. Coto, Social complexity and brain evolution: Insights from ant neuroarchitecture and genomics. *Curr. Opin. Insect Sci.* **53**, 100962 (2022).
41. R. I. M. Dunbar, The social brain hypothesis-thirty years on. *Ann. Hum. Biol.* **51**, 2359920 (2024).
42. O. Feinerman, J. F. A. Traniello, Social complexity, diet, and brain evolution: Modeling the effects of colony size, worker size, brain size, and foraging behavior on colony fitness in ants. *Behav. Ecol. Sociobiol.* **70**, 1063–1074 (2016).
43. J. M. DeSilva, J. F. A. Traniello, A. G. Claxton, L. D. Fannin, When and why did human brains decrease in size? A new change-point analysis and insights from brain evolution in ants. *Front. Ecol. Evol.* **9**, 742639 (2021).
44. J. M. DeSilva *et al.*, Human brains have shrunk: The questions are when and why. *Front. Ecol. Evol.* **11**, 1191274 (2023).
45. J. F. Kamhi, W. Gronenberg, S. K. Robson, J. F. A. Traniello, Social complexity influences brain investment and neural operation costs in ants. *Proc. Biol. Sci.* **283**, 2016949 (2016).
46. L. Chittka, J. Niven, Are bigger brains better? *Curr. Biol.* **19**, R995–R1008 (2009).
47. R. K. Godfrey, M. Swartzlander, W. Gronenberg, Allometric analysis of brain cell number in Hymenoptera suggests ant brains diverge from general trends. *Proc. Biol. Sci.* **288**, 20210199 (2021).
48. B. Sengupta, M. Stemmler, S. B. Laughlin, J. E. Niven, Action potential energy efficiency varies among neuron types in vertebrates and invertebrates. *PLoS. Comput. Biol.* **6**, e1000840 (2010).
49. J. E. Niven, S. M. Farris, Miniaturization of nervous systems and neurons. *Curr. Biol.* **22**, R323–R329 (2012).
50. J. E. Niven, Neuronal energy consumption: Biophysics, efficiency and evolution. *Curr. Opin. Neurobiol.* **41**, 129–135 (2016).
51. Z. N. Coto, J. F. A. Traniello, Brain size, metabolism and social evolution. *Front. Physiol.* **12**, 612865 (2021).
52. Z. N. Coto, J. F. A. Traniello, Social brain energetics: Ergonomic efficiency, neurometabolic scaling, and metabolic polyphenism in ants. *Integr. Comp. Biol.* **62**, 1471–1478 (2022).
53. J. S. Waters, J. F. Harrison, "Insect metabolic rates" in *Metabolic Ecology: A Scaling Approach*, R. M. Silby, J. H. Brown, A. Kodric-Brown, Eds. (John Wiley & Sons Ltd, 2012), pp. 198–211.
54. J. F. Harrison, Approaches for testing hypotheses for the hypometric scaling of aerobic metabolic rate in animals. *Am. J. Physiol. Regul. Integr, Comp. Physiol.* **315**, R879–894 (2018).
55. J. F. Harrison *et al.*, White paper: An integrated perspective on the causes of hypometric metabolic scaling in animals. *Integr. Comp. Biol.* **62**, 1395–1418 (2022).
56. J. Z. Shik, The metabolic costs of building ant colonies from variably sized subunits. *Behav. Ecol. Sociobiol.* **64**, 1981–1990 (2010).
57. R. Tomer, A. S. Denes, K. Tessmar-Raible, D. Arendt, Profiling by image registration reveals common origin of annelid mushroom bodies and vertebrate pallium. *Cell* **142**, 800–809 (2010).
58. R. K. Vander Meer, L. Morel, "Nestmate recognition in ants" in *Pheromone Communication in Social Insects*, R. K. Vander Meer, M. Breed, M. Winston, K. E. Espelie, Eds. (Westview Press, 1997), pp. 79–103.
59. P. d'Ettore, A. Lenoir, "Nestmate recognition" in *Ant Ecology*, L. Lach, C. L. Parr, K. L. Abbot, Eds. (Oxford University Press, 2009), pp. 194–209.
60. K. Sano, N. Bannon, M. J. Greene, Pavement ant workers (*Tetramorium caespitum*) assess cues coded in cuticular hydrocarbons to recognize conspecific and heterospecific non-nestmate ants. *J. Insect Behav.* **31**, 186–199 (2018).
61. M. Bey, R. Endermann, C. Raudies, J. Steinle, V. Nehring, Associative learning of non-nestmate cues improves enemy recognition in ants. *Curr. Biol.* **35**, 407–412 (2025).
62. E. Csata, A. Dussutour, Nutrient regulation in ants (Hymenoptera: Formicidae): A review. *Myrmecol. News* **29**, 111–124 (2019).
63. A. P. Cruz-Neto, F. Bozinovic, The relationship between diet quality and basal metabolic rate in endotherms: Insights from intraspecific analysis. *Physiol. Biochem. Zool.* **77**, 877–889 (2004).
64. M. Poxleitner *et al.*, Western diet increases brain metabolism and adaptive immune responses in a mouse model of amyloidosis. *J. Neuroinflam.* **21**, 129 (2024).
65. E. O. Wilson, Enemy specification in the alarm-recruitment system of an ant. *Science* **190**, 798–800 (1975).
66. A. N. Bubak, J. D. W. Yaeger, K. J. Renner, J. G. Swallow, M. J. Greene, Neuromodulation of nestmate recognition decisions by pavement ants. *PLoS One* **11**, e0166417 (2016).
67. H. Li-Byarlay, C. C. Rittschof, J. H. Massey, B. R. Pittendrigh, G. E. Robinson, Socially responsive effects of brain oxidative metabolism on aggression. *Proc. Natl. Acad. Sci. U.S.A.* **111**, 12533–12537 (2014).
68. C. C. Rittschof, H. J. Vekaria, J. H. Palmer, P. G. Sullivan, Brain mitochondrial bioenergetics change with rapid and prolonged shifts in aggression in the honey bee, Apis mellifera. *J. Exp. Biol.* **221**, jeb176917 (2018).
69. J. F. Kamhi, S. Arganda, C. S. Moreau, J. F. A. Traniello, Origins of aminergic regulation of behavior in complex insect social systems. *Front. Syst. Neurosci.* **11**, 74 (2017).
70. C. C. Rittschof, H. J. Vekaria, J. H. Palmer, P. G. Sullivan, Biogenic amines and activity levels alter the neural energetic response to aggressive social cues in the honey bee Apis mellifera. *J. Neurosci. Res.* **97**, 991–1003 (2019).
71. E. L. Arrese, J. L. Soulages, Insect fat body: Energy, metabolism and regulation. *Annu. Rev. Entomol.* **55**, 207–225 (2010).
72. A. A. Weger, C. C. Rittschof, The diverse roles of insulin signaling in insect behavior. *Front. Insect Sci.* **4**, 1360320 (2024).
73. C. C. Rittschof *et al.*, Neuromolecular responses to social challenge: Common mechanisms across mouse, stickleback fish, and honey bee. *Proc. Natl. Acad. Sci. U.S.A.* **111**, 17929–17934 (2014).
74. A. Peters *et al.*, The selfish brain: Competition for energy resources. *Neurosci. Biobehav. Rev.* **28**, 143–180 (2004).
75. A. Peters, B. Kubera, C. Hubold, D. Langemann, The selfish brain: Stress and eating behavior. *Front. Neurosci.* **5**, 74 (2011).
76. P. Skowronek, L. Wójcik, A. Strachecka, Fat body–Multifunctional insect tissue. *Insects* **12**, 547 (2021).
77. A. Rajan, N. Perrimon, Drosophila cytokine unpaired 2 regulates physiological homeostasis by remotely controlling insulin secretion. *Cell* **151**, 123–137 (2012).
78. J. Mattila, V. Hietakangas, Regulation of carbohydrate energy metabolism in Drosophila melanogaster. *Genetics* **207**, 1231–1253 (2017).
79. S. K. Kim, E. J. Rulifson, Conserved mechanisms of glucose sensing and regulation by Drosophila corpora cardiaca cells. *Nature* **431**, 316–320 (2004).
80. G. Isabel, J.-R. Martin, S. Chidami, J. A. Veenstra, P. Rosay, AKH-producing neuroendocrine cell ablation decreases trehalose and induces behavioral changes in *Drosophila*. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **288**, R531–R538 (2004).
81. S. Grönke *et al.*, Dual lipolytic control of body fat storage and mobilization in Drosophila. *PLoS Biol.* **5**, e137 (2007).
82. U. Toprak, The role of peptide hormones in insect lipid metabolism. *Front. Physiol.* **11**, 434 (2020).
83. S. Satake *et al.*, Bombyxin, an insulin-related peptide of insects, reduces the major storage carbohydrates in the silkworm *Bombyx mori*. *Comp. Biochem. Physiol. B Biochem. Mol. Biol.* **118**, 349–357 (1997).
84. M. S. Defferrari, I. Orchard, A. B. Lange, Identification of the first insulin-like peptide in the disease vector Rhodnius prolixus: Involvement in metabolic homeostasis of lipids and carbohydrates. *Insect Biochem. Mol. Biol.* **70**, 148–159 (2016).
85. S. Chowański *et al.*, Insulin-like peptides and cross-talk with other factors in the regulation of insect metabolism. *Front. Physiol.* **12**, 701203 (2021).
86. Y. Wang, O. Kaftanoglu, C. S. Brent, R. E. Page Jr., G. V. Amdam, Starvation stress during larval development facilitates an adaptive response in adult worker honey bees (Apis mellifera L.). *J. Exp. Biol.* **219**, 949–959 (2016).
87. V. Chandra *et al.*, Social regulation of insulin signaling and the evolution of eusociality in ants. *Science* **361**, 398–402 (2018).
88. H. Yan *et al.*, Insulin signaling in the long-lived reproductive caste of ants. *Science* **377**, 1092–1099 (2022).
89. W. Zhou *et al.*, Uncovering the function of insulin receptor substrate in termites' immunity through active immunization. *J. Insect Sci.* **24**, 1 (2024).
90. B. Silva *et al.*, Glia fuel neurons with locally synthesized ketone bodies to sustain memory under starvation. *Nat. Metab.* **4**, 213–224 (2022).
91. M. S. Aydin *et al.*, Active shrinkage protects neurons following axonal transection. *iScience* **26**, 107715 (2023).
92. R. K. Godfrey, W. Gronenberg, Brain evolution in social insects: Advocating for the comparative approach. *J. Comp. Physiol. A* **205**, 13–32 (2019).
93. S. Oikawa, Y. Itazawa, Relationship between metabolic rate in vitro and body mass in a marine teleost, porgy *Pagrus major*. *Fish Physiol. Biochem.* **10**, 177–182 (1992).

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.

94. S. Oikawa, Y. Itazawa, Relationship between summated tissue respiration and body size in a marine teleost, the porgy Pagrus major. *Fish. Sci.* **69**, 687–694 (2003).
95. J. J. Harris, R. Jolivet, D. Attwell, Synaptic energy use and supply. *Neuron* **75**, 762–777 (2012).
96. Y. Yu, P. Herman, D. L. Rothman, D. Agarwal, F. Hyder, Evaluating the gray and white matter energy budgets of human brain function. *J. Cereb. Blood Flow Metab.* **38**, 1339–1353 (2018).
97. C. Pulido, T. A. Ryan, Synaptic vesicle pools are a major hidden resting metabolic burden of nerve terminals. *Sci. Adv.* **7**, eabi9027 (2021).
98. J. F. Harrison, Do performance–safety tradeoffs cause hypometric metabolic scaling in animals? *Trends Ecol. Evol.* **32**, 653–664 (2017).
99. M. R. E. Symonds, M. A. Elgar, The evolution of body size, antennal size and host use in parasitoid wasps (Hymenoptera: Chalcidoidea): A phylogenetic comparative analysis. *PLoS One* **8**, e78297 (2013).
100. A. A. Makarova, A. A. Diakova, S. U. Chaika, A. A. Polilov, Scaling of the sense organs of insects.1. Introduction. Compound eyes. *Entomol. Rev.* **102**, 161–181 (2022).
101. A. A. Makarova, A. A. Diakova, S. U. Chaika, A. A. Polilov, Scaling of the sense organs of insects. 2. Sensilla. Discussion. Conclusion. *Entomol. Rev.* **102**, 323–346 (2022).
102. P. J. Simmons, Scaling of sense organs that control flight: Size and sensory cell number of dipteran fly halteres. *J. Zool.* **322**, 12–23 (2023).
103. C. R. White, L. A. Alton, C. L. Bywater, E. J. Lombardi, D. J. Marshall, Metabolic scaling is the product of life-history optimization. *Science* **377**, 834–839 (2022).
104. D. S. Glazier, Does death drive the scaling of life? *Biol. Rev. Camb. Philos. Soc.* **100**, 586–619 (2024).
105. M. Papadopoulou, S. Cassenaer, T. Nowotny, G. Laurent, Normalization for sparse encoding of odors by a wide-field interneuron. *Science* **332**, 721–725 (2011).
106. A. C. Lin, A. M. Bygrave, A. de Calignon, T. Lee, G. Misenböck, Sparse, decorrelated odor coding in the mushroom body enhances learned odor discrimination. *Nat. Neurosci.* **17**, 559–568 (2014).
107. A. D. Moss, J. G. Swallow, M. J. Greene, Always under foot: Tetramorium immigrans (Hymenoptera: Formicidae), a review. *Myrmecol. News* **32**, 75–92 (2022).
108. C. A. Penick, A. M. Savage, R. R. Dunn, Stable isotopes reveal links between human food inputs and urban ant diets. *Proc. Biol. Sci.* **282**, 20142608 (2015).
109. P. E. Schilman, F. Roces, Foraging energetics of a nectar-feeding ant: Metabolic expenditure as a function of food-source profitability. *J. Exp. Biol.* **209**, 4091–4101 (2006).
110. E. Nicholls, M. Rossi, J. E. Niven, Larval nutrition impacts survival to adulthood, body size and the allometric scaling of metabolic rate in adult honeybees. *J. Exp. Biol.* **224**, 242393 (2021).
111. S. Sanada-Morimura, T. Satoh, Y. Obara, Territorial behavior and temperature preference for nesting sites in a pavement ant Tetramorium tsushimae. *Insect. Soc.* **53**, 141–148 (2006).
112. J. R. B. Lighton, *Measuring Metabolic Rates: A Manual for Scientists* (Oxford University Press, 2019).
113. E. O. Wilson, *Pheidole in the New World. A Dominant, Hyperdiverse Ant Genus* (Harvard University Press, 2003).
114. E. O. Wilson, The organization of colony defense in the ant Pheidole dentata Mayr (Hymenoptera: Formicidae). *Behav. Ecol. Sociobiol.* **1**, 63–81 (1976).
115. D. Bates, M. Mächler, B. Bolker, S. Walker, Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* **67**, 1–48 (2015).
116. H. Wickham, *ggplot2: Elegant Graphics for Data Analysis* (Springer-Verlag, New York, 2016).
117. D. I. Warton, R. A. Duursma, D. S. Falster, S. Taskinen, Smatr 3–An R package for estimation and inference about allometric lines. *Methods Ecol. Evol.* **3**, 257–259 (2012).
118. R. Lenth, J. Piaskowski, emmeans: Estimated Marginal Means, aka Least-Squares Means (R package version 2.0.3, CRAN, 2026). https://rvlenth.github.io/emmeans/. Acessed 16 April 2026.

Downloaded from https://www.pnas.org by UNIVERSITY OF BATH on October 30, 2026 from IP address 138.38.93.65.